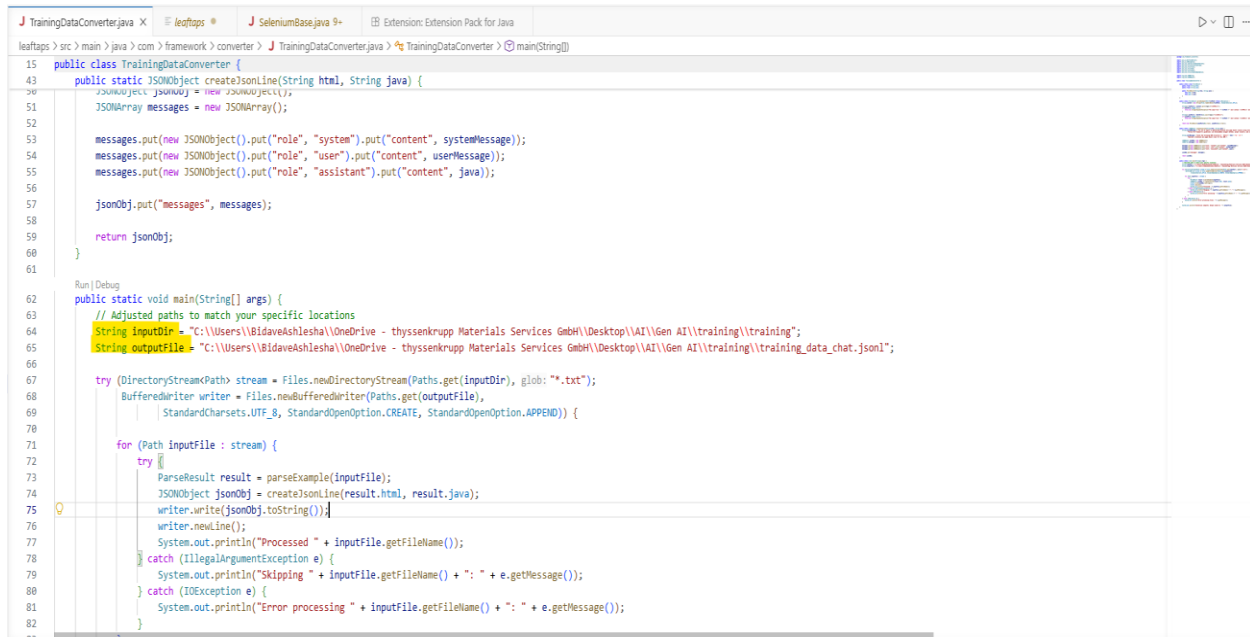


STEPS FOR FINE TUNING A MODEL

1.)Setting up the paths for input and output in java convertor file



```
15 public class TrainingDataConverter {
43     public static JSONObject createJsonLine(String html, String java) {
44         JSONObject jsonObj = new JSONObject();
51         JSONArray messages = new JSONArray();
52
53         messages.put(new JSONObject().put("role", "system").put("content", systemMessage));
54         messages.put(new JSONObject().put("role", "user").put("content", userMessage));
55         messages.put(new JSONObject().put("role", "assistant").put("content", java));
56
57         jsonObj.put("messages", messages);
58
59         return jsonObj;
60     }
61
62     public static void main(String[] args) {
63         // Adjusted paths to match your specific locations
64         String inputDir = "C:\\Users\\Bldavekshlesha\\OneDrive - thyssenkrupp Materials Services GmbH\\Desktop\\AI\\Gen AI\\training\\training";
65         String outputFile = "C:\\Users\\Bldavekshlesha\\OneDrive - thyssenkrupp Materials Services GmbH\\Desktop\\AI\\Gen AI\\training\\training_data_chat.jsonl";
66
67         try (DirectoryStream<Path> stream = Files.newDirectoryStream(Paths.get(inputDir), glob: "*.txt");
68              BufferedWriter writer = Files.newBufferedWriter(Paths.get(outputFile),
69                  StandardCharsets.UTF_8, StandardOpenOption.CREATE, StandardOpenOption.APPEND)) {
70
71             for (Path inputFile : stream) {
72                 try {
73                     ParseResult result = parseExample(inputFile);
74                     JSONObject jsonObj = createJsonLine(result.html, result.java);
75                     writer.write(jsonObj.toString());
76                     writer.newLine();
77                     System.out.println("Processed " + inputFile.getFileName());
78                 } catch (IllegalArgumentException e) {
79                     System.out.println("Skipping " + inputFile.getFileName() + ": " + e.getMessage());
80                 } catch (IOException e) {
81                     System.out.println("Error processing " + inputFile.getFileName() + ": " + e.getMessage());
82                 }
83             }
84         }
85     }
86 }
```

2.) Created create lead page

Name	Status	Date modified	Type	Size
 CreateLeadPage	🟢	5/28/2026 1:58 AM	TXT File	2 KB
 EditLeadPage	🟢	5/27/2026 10:19 AM	TXT File	2 KB
 FindLeadsPage	🟢	5/27/2026 10:19 AM	TXT File	3 KB
 FindLeadsUpdated	🟢	5/27/2026 10:19 AM	TXT File	3 KB
 HomePage	🟢	5/27/2026 10:19 AM	TXT File	2 KB
 Login	🟢	5/27/2026 10:19 AM	TXT File	2 KB
 MyHomePage	🟢	5/27/2026 10:19 AM	TXT File	1 KB
 MyLeadsPage	🟢	5/27/2026 10:19 AM	TXT File	2 KB
 OrderEntryHeaderPage	🟢	5/27/2026 10:19 AM	TXT File	5 KB
 UploadPage	🟢	5/27/2026 10:19 AM	TXT File	2 KB
 ViewLeadPage	🟢	5/27/2026 10:19 AM	TXT File	3 KB

3.) Run the program

```
J TrainingDataConverter.java X leafaps SeleniumBase.java 9+ Extension: Extension Pack for Java
leafaps > src > main > java > com > framework > converter > J TrainingDataConverter.java > TrainingDataConverter > main(String[])

15 public class TrainingDataConverter {
43     public static JSONObject createJsonLine(String html, String java) {
50         JSONObject jsonObj = new JSONObject();
51         JSONArray messages = new JSONArray();
52
53         messages.put(new JSONObject().put("role", "system").put("content", systemMessage));
54         messages.put(new JSONObject().put("role", "user").put("content", userMessage));
55         messages.put(new JSONObject().put("role", "assistant").put("content", java));
56
57         jsonObj.put("messages", messages);
58
59         return jsonObj;
60     }
61
62     Run | Debug
63     public static void main(String[] args) {
64         // Adjusted paths to match your specific locations
65         String inputDir = "C:\\Users\\Bidaresha\\OneDrive - thyssenkrupp Materials Services GmbH\\Desktop\\AI\\Gen AI\\training\\training";
66         String outputFile = "C:\\Users\\Bidaresha\\OneDrive - thyssenkrupp Materials Services GmbH\\Desktop\\AI\\Gen AI\\training\\training_data_chat.jsonl";
67
68         try (DirectoryStream stream = Files.newDirectoryStream(Paths.get(inputDir), glob: "**.txt");
69             BufferedWriter writer = Files.newBufferedWriter(Paths.get(outputFile),
70                 StandardCharsets.UTF_8, StandardOpenOption.CREATE, StandardOpenOption.APPEND)) {
71
72             for (Path inputFile : stream) {
73                 try {
74                     ParseResult result = parseExample(inputFile);
75                     JSONObject jsonObj = createJsonLine(result.html, result.java);
76                     writer.write(jsonObj.toString());
77                     writer.newLine();
78                     System.out.println("Processed " + inputFile.getFileName());
79                 } catch (IllegalArgumentException e) {
80                     System.out.println("Skipping " + inputFile.getFileName() + ": " + e.getMessage());
81                 } catch (IOException e) {
82                     System.out.println("Error processing " + inputFile.getFileName() + ": " + e.getMessage());
83                 }
84             }
85         }
86     }
87 }

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

● PS C:\Users\Bidaresha\OneDrive - thyssenkrupp Materials Services GmbH\Desktop\AI\Gen AI\leafaps\leafaps> cd 'C:\Users\Bidaresha\OneDrive - thyssenkrupp Materials Services GmbH\Desktop\AI\Gen AI\leafaps\leafaps'; & 'C:\Sof
re\jdk-20.0.2\bin\java.exe' '@C:\Users\BIDAVE-1\AppData\Local\Temp\cp_7r17168gcknpg62ofjlbqkrh.argfile' 'com.framework.converter.TrainingDataConverter'
Conversion complete. Output saved to: C:\Users\Bidaresha\OneDrive - thyssenkrupp Materials Services GmbH\Desktop\AI\Gen AI\training\training_data_chat.jsonl
● PS C:\Users\Bidaresha\OneDrive - thyssenkrupp Materials Services GmbH\Desktop\AI\Gen AI\leafaps\leafaps> cd 'C:\Users\Bidaresha\OneDrive - thyssenkrupp Materials Services GmbH\Desktop\AI\Gen AI\leafaps\leafaps'; & 'C:\Sof
re\jdk-20.0.2\bin\java.exe' '@C:\Users\BIDAVE-1\AppData\Local\Temp\cp_7r17168gcknpg62ofjlbqkrh.argfile' 'com.framework.converter.TrainingDataConverter'
Processed EditLeadPage.txt
Processed FindLeadPage.txt
Processed FindLeadsUpdated.txt
Processed HomePage.txt
Processed Login.txt
Processed MyHomePage.txt
Processed MyLeadsPage.txt
Processed OrderEntryHeaderPage.txt
Processed UploadPage.txt
Processed ViewLeadPage.txt
Conversion complete. Output saved to: C:\Users\Bidaresha\OneDrive - thyssenkrupp Materials Services GmbH\Desktop\AI\Gen AI\training\training_data_chat.jsonl
● PS C:\Users\Bidaresha\OneDrive - thyssenkrupp Materials Services GmbH\Desktop\AI\Gen AI\leafaps\leafaps> cd 'C:\Users\Bidaresha\OneDrive - thyssenkrupp Materials Services GmbH\Desktop\AI\Gen AI\leafaps\leafaps'; & 'C:\Sof
re\jdk-20.0.2\bin\java.exe' '@C:\Users\BIDAVE-1\AppData\Local\Temp\cp_7r17168gcknpg62ofjlbqkrh.argfile' 'com.framework.converter.TrainingDataConverter'
Processed CreateLeadPage.txt
Processed EditLeadPage.txt
Processed FindLeadPage.txt
Processed FindLeadsUpdated.txt
Processed HomePage.txt
Processed Login.txt
Processed MyHomePage.txt
Processed MyLeadsPage.txt
Processed OrderEntryHeaderPage.txt
Processed UploadPage.txt
Processed ViewLeadPage.txt
Conversion complete. Output saved to: C:\Users\Bidaresha\OneDrive - thyssenkrupp Materials Services GmbH\Desktop\AI\Gen AI\training\training_data_chat.jsonl
● PS C:\Users\Bidaresha\OneDrive - thyssenkrupp Materials Services GmbH\Desktop\AI\Gen AI\leafaps\leafaps> cd 'C:\Users\Bidaresha\OneDrive - thyssenkrupp Materials Services GmbH\Desktop\AI\Gen AI\leafaps\leafaps'; & 'C:\Sof
re\jdk-20.0.2\bin\java.exe' '@C:\Users\BIDAVE-1\AppData\Local\Temp\cp_7r17168gcknpg62ofjlbqkrh.argfile' 'com.framework.converter.TrainingDataConverter'
```

4.) jsonl file is created

```
J TrainingDataConverter.java training_data_chat.jsonl X leafaps SeleniumBase.java 9+ Extension: Extension Pack for Java
C:\Users\Bidaresha\OneDrive - thyssenkrupp Materials Services GmbH\Desktop\AI\Gen AI\training> .\training_data_chat.jsonl

1 ["messages":[{"role":"system","content":"You are an expert in generating Selenium Java page object classes using provided DOM information. Follow the framework's guidelines: use SeleniumBase wrapper methods, proper
2 ["messages":[{"role":"system","content":"You are an expert in generating Selenium Java page object classes using provided DOM information. Follow the framework's guidelines: use SeleniumBase wrapper methods, proper
3 ["messages":[{"role":"system","content":"You are an expert in generating Selenium Java page object classes using provided DOM information. Follow the framework's guidelines: use SeleniumBase wrapper methods, proper
4 ["messages":[{"role":"system","content":"You are an expert in generating Selenium Java page object classes using provided DOM information. Follow the framework's guidelines: use SeleniumBase wrapper methods, proper
5 ["messages":[{"role":"system","content":"You are an expert in generating Selenium Java page object classes using provided DOM information. Follow the framework's guidelines: use SeleniumBase wrapper methods, proper
6 ["messages":[{"role":"system","content":"You are an expert in generating Selenium Java page object classes using provided DOM information. Follow the framework's guidelines: use SeleniumBase wrapper methods, proper
7 ["messages":[{"role":"system","content":"You are an expert in generating Selenium Java page object classes using provided DOM information. Follow the framework's guidelines: use SeleniumBase wrapper methods, proper
8 ["messages":[{"role":"system","content":"You are an expert in generating Selenium Java page object classes using provided DOM information. Follow the framework's guidelines: use SeleniumBase wrapper methods, proper
9 ["messages":[{"role":"system","content":"You are an expert in generating Selenium Java page object classes using provided DOM information. Follow the framework's guidelines: use SeleniumBase wrapper methods, proper
10 ["messages":[{"role":"system","content":"You are an expert in generating Selenium Java page object classes using provided DOM information. Follow the framework's guidelines: use SeleniumBase wrapper methods, proper
11 ["messages":[{"role":"system","content":"You are an expert in generating Selenium Java page object classes using provided DOM information. Follow the framework's guidelines: use SeleniumBase wrapper methods, proper
12 ["messages":[{"role":"system","content":"You are an expert in generating Selenium Java page object classes using provided DOM information. Follow the framework's guidelines: use SeleniumBase wrapper methods, proper
```

5.) Beautify the create lead page from Ilm

beautify the json

```
{
  "messages": [
    {
      "role": "system",
      "content": "You are an expert in generating Selenium Java page object classes using provided DOM information. Follow the framework's guidelines: use SeleniumBase wrapper methods, proper locators, and insert TODO comments if a mapping is missing. Make sure every page method includes comments in your generated code as the training examples may not have comments",
    },
    {
      "role": "user",
      "content": "Given the following DOM structure:\n<html>\n<input type='text' class='inputBox' name='firstName' size='30' maxLength='100'"
    }
  ]
}
```

Show more ▾

JSON

```
{
  "messages": [
    {
      "role": "system",
      "content": "You are an expert in generating Selenium Java page object classes using provided DOM information. Follow the framework's guidelines: use SeleniumBase wrapper methods, proper locators, and insert TODO comments if a mapping is missing. Make sure every page method includes comments in your generated code as the training examples may not have comments",
    },
    {
      "role": "user",
      "content": "Given the following DOM structure:\n<html>\n<input type='text' class='inputBox' name='firstName' size='30' maxLength='100'",
    },
    {
      "role": "assistant",
      "content": "package com.leafyaps.pages;\n\nimport com.framework.selenium.api.design.Locators;\nimport com.framework.selenium.api.design.Page;\n\npublic class LoginPage {\n\n    // TODO: Add locators for the login page\n\n    // TODO: Add page methods for the login page\n\n}\n"
```



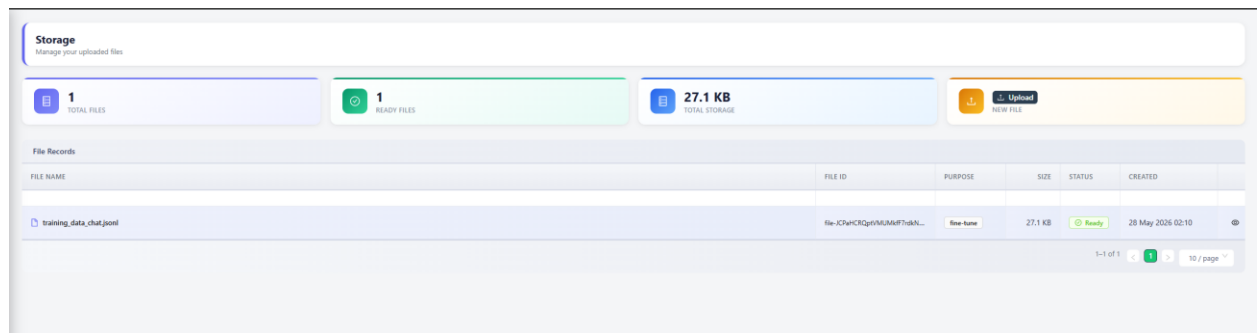
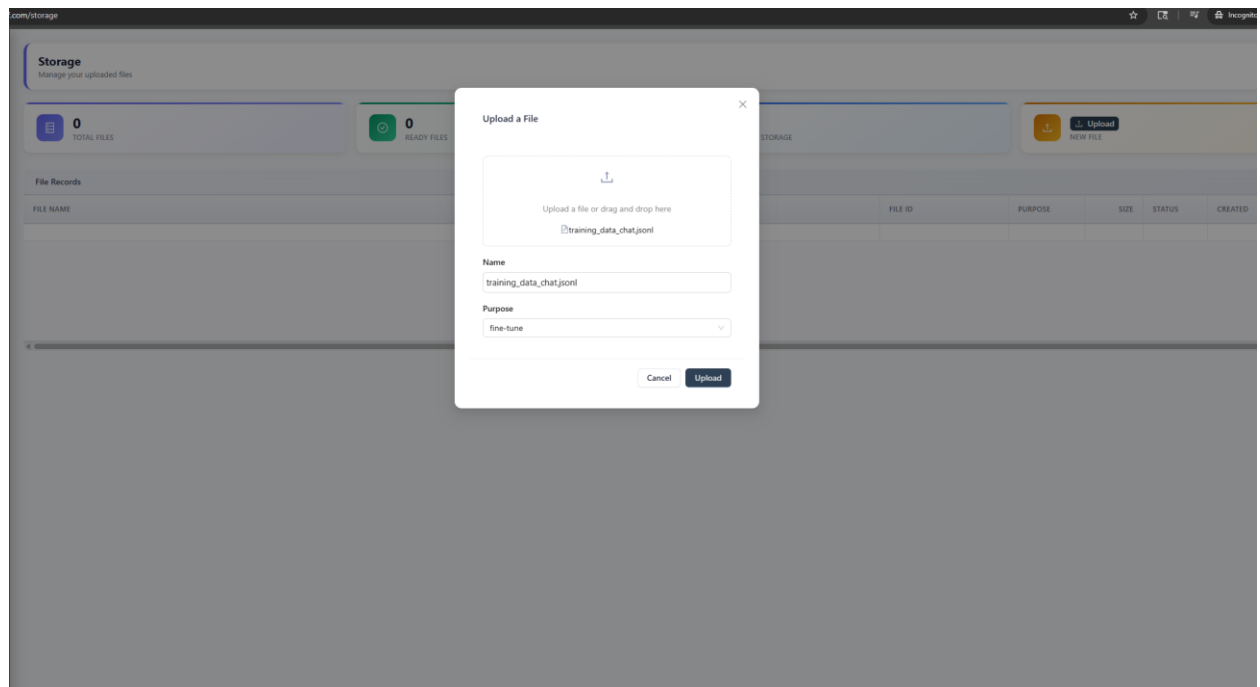
+ Ask anything



Voice

ChatGPT can make mistakes. Check important info.

6.) Logged into the testleaf.api > stored the jsonl file create in step 4



7.) Creating a Fine tuned Model

Create a fine-tuned model

METHOD

Supervised

TRAINING DATA

Upload new

Select existing

file-JCPaHCRQptVMUMkfF7rdkN

BASE MODEL

gpt-4o-mini-2024-07-18

Base model is required.

VALIDATION DATA

Upload new

Select existing

None

SEED

Controls reproducibility. Same seed + parameters produces the same result. Leave 0 to auto-generate.

0

HYPERPARAMETERS

Batch Size

1

Learning Rate Multiplier

0.61

Recommended range: 0.0001 – 10

Number of Epochs

4

Recommended range: 1 – 10

Cancel

Create Model

8.) Fining tuning process kickstarted

Fine-tuning

Train and manage custom models

1

TOTAL JOBS

0

SUCCEEDED

0

FAILED

0

TRAINED TOKENS

All

Successful

Failed

Create

1-1 of 110 / page

Fine-tuning

Train and manage custom models

1

TOTAL JOBS

0

SUCCEEDED

0

FAILED

0

TRAINED TOKENS

All

Successful

Failed

Create

1-1 of 110 / page

9.) Modified the model in browser extension


```

import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;

/**
 * Page Object for Salesforce Login Page
 */
public class SalesforceLoginPage {
    private WebDriver driver;

    // Locators
    private By usernameField = By.id("username");
    private By passwordField = By.id("password");
    private By loginButton = By.id("Login");
    private By rememberMeCheckbox = By.id("rememberUn");
    private By forgotPasswordLink = By.id("forgot_password_link");
    private By customDomainLink = By.id("mydomainLink");

    /**
     * Constructor for SalesforceLoginPage
     *
     * @param driver WebDriver instance
     */
    public SalesforceLoginPage(WebDriver driver) {
        this.driver = driver;
    }

    /**

```

```

* Enters the username in the username field.
*
* @param username the username to enter
*/
public void enterUsername(String username) {
    WebElement usernameElement = driver.findElement(usernameField);
    usernameElement.clear();
    usernameElement.sendKeys(username);
}

/**
* Enters the password in the password field.
*
* @param password the password to enter
*/
public void enterPassword(String password) {
    WebElement passwordElement = driver.findElement(passwordField);
    passwordElement.clear();
    passwordElement.sendKeys(password);
}

/**
* Clicks the login button to submit the login form.
*/
public void clickLoginButton() {
    driver.findElement(loginButton).click();
}

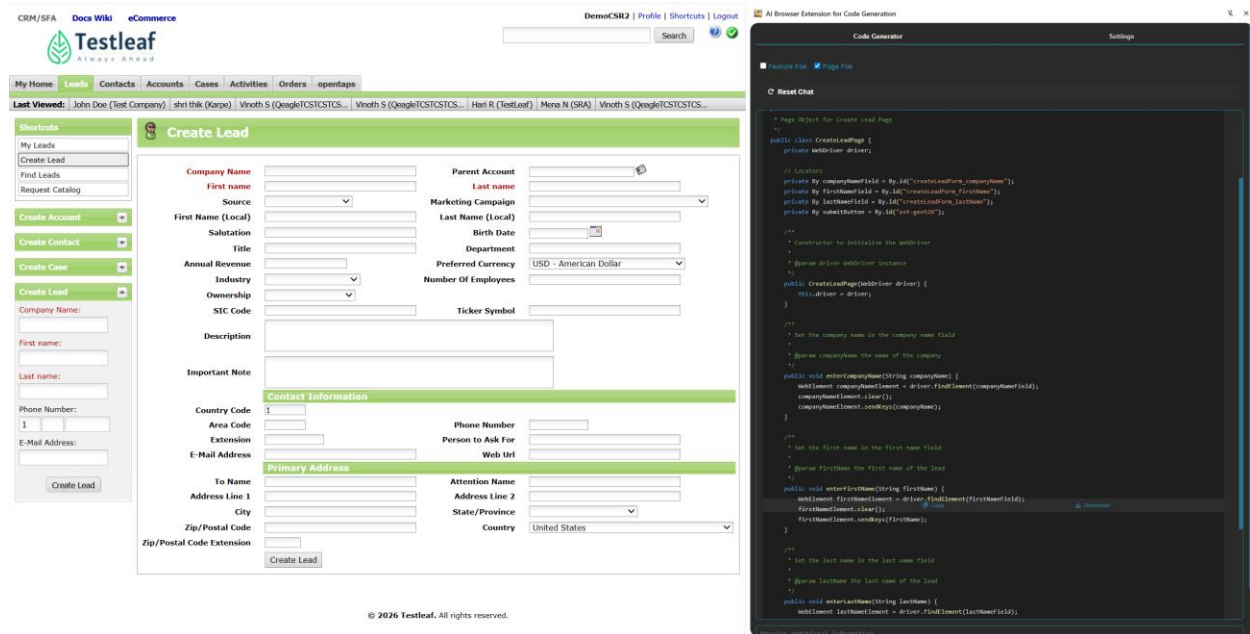
```

```
/**
 * Checks or unchecks the "Remember me" checkbox.
 *
 * @param remember true to check the checkbox, false to uncheck
 */
public void setRememberMe(boolean remember) {
    WebElement checkbox = driver.findElement(rememberMeCheckbox);
    if (checkbox.isSelected() != remember) {
        checkbox.click();
    }
}
```

```
/**
 * Clicks the "Forgot Your Password?" link.
 */
public void clickForgotPasswordLink() {
    driver.findElement(forgotPasswordLink).click();
}
```

```
/**
 * Clicks the "Use Custom Domain" link.
 */
public void clickCustomDomainLink() {
    driver.findElement(customDomainLink).click();
}
```

12.) Also tried for create lead



package com.testleaf.pages;

import org.openqa.selenium.By;

import org.openqa.selenium.WebDriver;

import org.openqa.selenium.WebElement;

/**

* Page Object for Create Lead Page

*/

public class CreateLeadPage {

private WebDriver driver;

// Locators

private By companyNameField = By.id("createLeadForm_companyName");

```
private By firstNameField = By.id("createLeadForm_firstName");
private By lastNameField = By.id("createLeadForm_lastName");
private By submitButton = By.id("ext-gen528");
```

```
/**
```

```
 * Constructor to initialize the WebDriver
```

```
 *
```

```
 * @param driver WebDriver instance
```

```
 */
```

```
public CreateLeadPage(WebDriver driver) {
    this.driver = driver;
}
```

```
/**
```

```
 * Set the company name in the company name field
```

```
 *
```

```
 * @param companyName the name of the company
```

```
 */
```

```
public void enterCompanyName(String companyName) {
    WebElement companyNameElement =
driver.findElement(companyNameField);
    companyNameElement.clear();
    companyNameElement.sendKeys(companyName);
}
```

```
/**
```

```
 * Set the first name in the first name field
```

```

*

* @param firstName the first name of the lead
*/

public void enterFirstName(String firstName) {
    WebElement firstNameElement = driver.findElement(firstNameField);
    firstNameElement.clear();
    firstNameElement.sendKeys(firstName);
}

/**
* Set the last name in the last name field
*
* @param lastName the last name of the lead
*/

public void enterLastName(String lastName) {
    WebElement lastNameElement = driver.findElement(lastNameField);
    lastNameElement.clear();
    lastNameElement.sendKeys(lastName);
}

/**
* Click the submit button to create the lead
*/

public void clickSubmitButton() {
    WebElement submitButtonElement = driver.findElement(submitButton);
    submitButtonElement.click();
}

```

}

13.) Fine Tuned Succeeded

Fine-tuning

Train and manage custom models

1

TOTAL JOBS

1

SUCCEEDED

0

FAILED

23,284

TRAINED TOKENS

AllSuccessfulFailed

Create

FINE-TUNE ID	BASE MODEL	METHOD	STATUS	TRAINED TOKENS	CREATED	
ft:gpt-4o-mini-2024-07-18:testleaf-2:DkFX7Say	gpt-4o-mini-2024-07-18	supervised	Succeeded	23,284	28 May 2025 02:14	

1-1 of 1

123

10 / page

14.) updated in pop up json

```
src > scripts > JS popup.js > document.addEventListener('DOMContentLoaded') callback > updateModelOptions
5 document.addEventListener('DOMContentLoaded', async () => {
96 console.error('Chrome storage is not available');
97 return;
98 }
99
100 const providerSelect = document.getElementById('providerSelect');
101 const modelSelect = document.getElementById('modelSelect');
102 const groqApiKeyInput = document.getElementById('groqApiKey');
103 const openaiApiKeyInput = document.getElementById('openaiApiKey');
104 const testleafApiKeyInput = document.getElementById('testleafApiKey');
105
106 // Model options by provider
107 const modelsByProvider = {
108   groq: [
109     { value: 'deepseek-r1-distill-llama-70b', label: 'deepseek-r1-distill-llama-70b' },
110     { value: 'llama-3.3-70b-versatile', label: 'llama-3.3-70b-versatile' },
111   ],
112   openai: [
113     { value: 'openai/gpt-oss-120b', label: 'openai/gpt-oss-120b' },
114     { value: 'openai/gpt-oss-20b', label: 'openai-20b' },
115   ],
116   testleaf: [
117     { value: 'ft:gpt-4o-mini-2024-07-18:testleaf-2:DHArcAyu', label: 'Testleaf 2.0 Fine-tuned' },
118     { value: 'ft:gpt-4o-mini-2024-07-18:testleaf-2:DI2LHRx7', label: 'Testleaf 1.0' },
119     { value: 'ft:gpt-4o-mini-2024-07-18:testleaf-2:DkFX7Say', label: 'Testleaf 3.0' }
120   ]
121 };
122
123 // Function to update model options based on selected provider
124 function updateModelOptions(provider) {
125   modelSelect.innerHTML = `<option value="" disabled>Select the model...</option>`;
126
127   if (provider) {
128     modelSelect.disabled = false;
129     const models = modelsByProvider[provider] || [];
130     models.forEach(model => {
131       const option = document.createElement('option');
132       option.value = model.value;
133       option.textContent = model.label;
134       modelSelect.appendChild(option);
135     });
136   } else {
137     modelSelect.disabled = true;
138   }
139 }
140
141 // Handle provider selection
142 if (providerSelect) {
143   providerSelect.addEventListener('change', (e) => {
144     const provider = e.target.value;
145     storage.set({ selectedProvider: provider });
146     updateModelOptions(provider);
147     modelSelect.value = ''; // Reset model selection
148     storage.set({ selectedModel: '' }); // Clear stored model selection
149     updateApiKeyVisibility(''); // Hide API key inputs until model is selected
150   });
151 }
152
153 // Save API keys when entered
154 if (groqApiKeyInput) {
155   groqApiKeyInput.addEventListener('change', (e) => {
```


15.) 3.0 Model is displayed in ai extension

